METRIC

MIL-L-24773(SH)
24 March 1992

MILITARY SPECIFICATION

# LITHIUM BROMIDE ANHYDROUS AND LITHIUM BROMIDE BRINE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers lithium bromide for use in lithium bromide air conditioning plants.

1.2 Classification. The lithium bromide shall be of the following types, as specified (see 6.2):

Type I formulation - Lithium Bromide Anhydrous
Type II formulation - Lithium Bromide Brine

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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PPP-P-704 - Pails, Metal: (Shipping, Steel; 1 through 12 Gallons).

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible
MIL-D-43703 - Drums, Shipping and Storage, Molded Polyethylene

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished To Government Activities.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

(Unless otherwise indicated, copies of the federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of DoDISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 511 - Calcium and Magnesium in Water, Test Methods for. (DOD Adopted)
D 516 - Sulfate Ion In Water, Test Methods for Industrial Waste Water.
D 1067 - Acidity or Alkalinity of Water, Test Methods for. (DOD Adopted)
D 1068 - Test Method for Photometric Bathophenanthroline for BPA - Reactive and HCl - Reactive Ferrous Forms.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Type I formulation.

3.2.1 Material. The powdered lithium bromide anhydrous shall conform to the requirements specified herein and shall be free of all chemical inhibitors (additives).

3.2.2 Chemical properties. The constituents shall be in accordance with table I for type I formulation.

Table I. Chemical composition for type I formulation.

| Constituent Percent by Weight | Permissible concentration in formulation I | | Inspection |
|---|---|---|---|
| | Maximum amount | Minimum amount | |
| Chloride | 0.15 | ---- | ---- |
| Calcium | 0.01 | ---- | 4.6.1.1 |
| Magnesium | 0.001 | ---- | 4.6.1.1 |
| Iron | 0.002 | ---- | 4.6.1.2 |
| Sulfate | 0.02 | ---- | 4.6.1.3 |
| Alkalinity as $Li_2CO_3$ | 0.04 | ---- | 4.6.1.5 |
| Moisture | 1.0 | ---- | 4.6.1.6 |
| Water Insoluble | 0.05 | ---- | 4.6.1.7 |
| Lithium Bromide (LiBr) Content | ---- | 99.0 | 4.6.1.8 |

3.2.3 Physical properties.

3.2.3.1 Typical bulk density. The typical bulk density of the lithium bromide shall be 1.8 +/- 0.03 gram per milliliter (g/ml) when tested as specified in 4.6.2.1.

3.3 Type II formulation.

3.3.1 Material. The lithium bromide brine shall be a clear, colorless solution which shall conform to the requirements specified herein and shall be free of all chemical inhibitors (additives).



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3.3.2 Chemical properties. The constituents shall be in accordance with table II for Type II formulation.

Table II. Chemical composition for type II formulation.

| Constituent | Permissible concentration in formulation II | | Inspection |
|---|---|---|---|
| | Maximum amount | Minimum amount | |
| Chlorine (ppm) | 800.0 | ---- | ---- |
| Calcium (ppm) | 46.0 | ---- | 4.7.1.1 |
| Magnesium (ppm) | 5.0 | ---- | 4.7.1.1 |
| Iron (ppm) | 20.0 | ---- | 4.7.1.2 |
| Sulfate (ppm) | 200.0 | ---- | 4.7.1.3 |
| Lithium Bromide Content (weight percent) | 54.0 | 52.0 | 4.7.1.5 |
| Alkalinity (normality) | 0.105N | 0.090N | 4.7.1.6 |
| Water insoluble (ppm) | 200.0 | ---- | 4.7.1.7 |

3.3.3 Physical properties.

3.3.3.1 Specific gravity. The specific gravity of the lithium bromide shall be between 1.56 and 1.60 at 25 degrees Celsius (°C) (77 degrees Fahrenheit (°F)) when tested as specified in 4.7.2.1.

3.3.4 Material safety data sheet (MSDS). The contracting activity shall be provided with a Material Safety Data Sheet at the time of the contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of material covered by this specification (see 6.6).

3.5 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on two samples of the formulation being tested and shall consist of the examinations and tests specified in 4.6 and 4.7 for type I and type II formulations, respectively (see 6.4). In the event that a first article sample does not conform to any applicable requirements, the contractor shall investigate and correct the root cause of the discrepancy, revise associated process instructions, and retest samples to determine the effectiveness of corrective action.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on every lot submitted to the Government, and shall consist of the examination specified in 4.5 and the examinations and tests specified in 4.6 and 4.7 for type I and type II formulations, respectively (see 6.3).

4.4.1 Lot. For the purpose of quality conformance inspection sampling, a lot shall consist of lithium bromide prepared in one continuous, homogenous batch of raw materials, without any addition or change in composition or processing procedure. The introduction of any ingredient or process change shall constitute a new lot.



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4.4.2 Sampling for examination of filled containers. As a minimum, the contractor shall select in accordance with table III, a sample quantity of filled containers from each lot and inspect them in accordance with 4.5. Detection of any nonconforming characteristics or weight shall result in rejection of the entire lot. The contractor has the option of correcting the discrepancy, reinspecting, and resubmitting a conforming lot, or submitting a new lot which shall be inspected as specified herein.

Table III. Sampling for examination of filled containers.

| Lot size (number) of containers | Sample size (containers) |
|---|---|
| up to 3 | all |
| 4 to 25 | 3 |
| 26 to 50 | 5 |
| 51 to 90 | 6 |
| 91 to 150 | 7 |
| 151 to 280 | 10 |
| 281 to 500 | 11 |
| 501 to 1200 | 15 |
| 1201 to 3200 | 18 |
| 3201 to 10,000 | 22 |

4.4.3 Sampling for chemical and physical properties. From each lot of type I and type II formulation, the first, middle and last containers filled shall be selected. From each selected container of type I formulation, a one pound sample shall be drawn, placed in separate, clean, dry containers, and labeled to identify the lot represented. From each selected container of type II formulation, a two pound sample shall be drawn, placed in an airtight container, and labeled to identify the lot represented. Each sample shall be subjected to the tests specified in 4.6 for type I formulation and 4.7 for type II formulation. The balance of the samples shall be reserved for possible retest until acceptance by the Government. The contractor shall maintain for a period of three years after contract completion, records of inspections, tests, and resulting rejections and corrective action.

4.4.4 Nonconformance. If a nonconformance is found in any sample, the entire lot shall be rejected. The contractor has the option of screening, correcting, and retesting 100% of the lot or submitting a new lot which shall be tested as required herein.

4.5 Examination of filled containers. Each container selected in accordance with 4.4.2 shall be examined for defects of the containers and the closure, for evidence of leakage, and for unsatisfactory markings. Each container selected shall be measured to determine the volume of the contents. Type I and type II formulations shall be weighed to determine conformance to 5.2.1 and 5.2.2, respectively.



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4.6 Inspection test procedures for type I formulation.

4.6.1 Chemical properties.

4.6.1.1 Calcium and magnesium content. The calcium and magnesium content shall be determined by the atomic absorption spectrophotometric method in accordance with ASTM D 511, method B. The calcium and magnesium concentrations shall be as specified in table I.

4.6.1.2 Iron content. The iron content shall be determined by the atomic absorption spectrophotometric method in accordance with ASTM D 1068. The iron concentration shall be as specified in table I.

4.6.1.3 Sulfate content. The sulfate analysis shall be made in accordance with ASTM D 516, method A, the gravimetric method. Sulfate content shall be as specified in table I.

4.6.1.4 Instrumental Analysis. For Ca, Mg, Fe, and $SO_4$ using Inductively Coupled Plasma/Atomic Emission Spectrometry (ICP). The LiBr materials are analyzed by ICP using matrix matched samples and standards containing an internal standard solution. The instrument should be operated at 1200 watts forward power. The following analytical lines shall be used:

Ca - 3933.66 - Analyte
Mg - 2790.79 - Analyte
Fe - 2599.40 - Analyte
S - 1820.40 - Analyte

Cd - 2265.10 - Internal Standard
Y - 3710.30 - Internal Standard

Table IV shows analyte and internal standard pair ratios for type I formulation.

Table IV. Analyte and internal standard pair ratios (type I formulation).

| | |
|---|---|
| Ca to Y | Fe to Cd |
| Mg to Y | S to Cd |

Samples and standards are adjusted to contain 0.5 gram (g) of Li/100 milliliter (ml) and 10 ppm Y and Cd. Samples should be prepared by weighing 6.25 g LiBr (anhydrous) and transferring to a 100 ml volumetric flask containing the Y and Cd internal standard plus 1.0 ml of concentrated HCl. A blank is prepared using the matrix matching material containing 0.5 g of Li, the internal standards and 1 ml of concentrated HCl. Calculate analyte as follows:

$$\text{Analyte, ppm} = \frac{\text{ICP analyte [mg/l] - ICP blank [mg/l x 100}}{\text{Sample weight (g)}}$$

4.6.1.5 Alkalinity. The alkalinity as $Li_2CO_3$ shall be as specified in table I. The alkalinity of the sample shall be determined in accordance with ASTM D 1067, method B. A 10 gram sample is dissolved in 100 ml of freshly boiled



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distilled water. The solution is titrated with 0.02 N hydrochloric acid using phenolphthalein as the indicator. The milli-equivalent weight (meg wt) for $Li_2CO_3$, the milliliters of titrant, the titrant normality, and the sample weight are used to calculate the percent by weight alkalinity expressed as equivalent weight (n) of normal $Li_2CO_3$.

$$\text{Alkalinity (as } Li_2CO_3\text{), percent by weight} = \frac{\text{Milliliters of titrant (ml) x 0.02 N x 0.037g/meg wt x 100}}{\text{Sample weight (g)}}$$

4.6.1.6 Moisture content. A sample of approximately 10 g shall be placed in a tared weighing bottle without unnecessary exposure to air and weighed to the nearest milligram. The sample shall be dried at 120°C (250°F) for four hours with the cover removed. At the conclusion of the drying period, the LiBr shall be placed in a desiccator to cool. After cooling to room temperature, the bottle containing the LiBr shall be weighed again, and the weight of the dried material determined. The moisture content of the original LiBr shall be determined in percent from the difference in weight of the material before and after drying, and shall be as specified in table I.

$$\text{Water, percent} = \frac{[(\text{Original mass of weighing bottle + sample) (g)} - (\text{Dried mass of weighing bottle + sample) (g)}] \text{ x } 100}{\text{(Original mass of sample) (g)}}$$

4.6.1.6.1 Optional moisture content. Weigh accurately a dry 50 ml glass-stoppered Erlenmeyer flask, using a similar flask as a counterpoise. Fold a piece of glossy paper into a funnel and insert it into the neck of the flask. Quickly transfer 3 to 5 g of the sample to the flask, discard the funnel and insert the stopper at once. Weigh the flask and contents, momentarily opening the counterpoise flask before weighing, to equalize the air pressure. Place the flask in a drying oven at 350°C (662°F), laying the stopper across the neck of the flask. Heat for at least 5 hours. Remove the flask from the oven, replace the stopper and cool in a desiccator for 1-1/2 hours. Weigh the flask, opening both the sample flask and counterpoise flask momentarily before weighing. Calculate moisture content as follows:

$$\text{Water, percent} = \frac{\text{Loss in weight (g) x 100}}{\text{Sample weight (g)}}$$

4.6.1.7 Water insoluble. Weigh a 10 g sample of the lithium bromide to the nearest 0.1g, transfer to a 250 ml beaker and dissolve with 150 ml of water. Warm the solution on a hot plate and digest for 15 minutes. Filter the solution through a dried, weighed Gooch crucible fitted with an asbestos mat, or a fiber glass filter disc, using a filter flask and suction. After transferring the insoluble quantitatively, wash the filter and insoluble thoroughly with jot water. Dry the crucible in an oven at 110°C (230°F) for one hour, cool in a desiccator and weigh.



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Water insoluble, percent =

$$\frac{\text{Weight of dried residue (g) x 100}}{\text{Sample weight (g)}}$$

The water insoluble (suspended solids) content shall be as specified in table I.

4.6.1.8 Lithium bromide content. The bromide ion concentration is determined by a gravimetric method. The bromide ion content is weighed as silver bromide and then calculated as an equivalent amount of lithium bromide. Dissolve 0.400 g of lithium bromide sample in 250 ml of distilled water. While stirring, gradually add 5 percent solution of silver nitrate, acidified with 1.0 ml of 1:99 nitric acid, until the precipitate coagulates. Heat the solution to 60°C (140°F) and let the precipitate settle; then add silver nitrate solution one drop at a time until the precipitation is complete. Set the flask aside for at least two hours, or preferably overnight. Pour the solution through a tared Gooch crucible. Wash all the precipitate from the beaker into the crucible with 0.05 grams per liter (g/l) silver nitrate solution. Wash the precipitate with the minimum amount of 1:99 nitric acid needed to remove excess silver nitrate, and then wash twice with distilled water to remove the acid. Dry the crucible and its contents at 100°C (212°F) for one hour. Cool in a desiccator and weigh. Repeat heating and drying until successive weighings agree within 0.5 mg. Calculate the percent by weight of lithium bromide as follows:

Lithium Bromide, percent by weight =

$$\frac{\text{46.25 x weight of precipitate (g)}}{\text{Sample weight (g)}}$$

4.6.1.8.1 Optional lithium bromide content. Determine the weight percent each of water, $Li_2CO_3$, Cl, $SO_4$, Ca, Mg, Fe, and Insoluble. Calculate percent by weight of lithium bromide as follows:

Percent (%) Total Impurities =

$$\% H_2O + \% Li_2CO_3 + \% Cl + \% SO_4 + \% Ca \; \% Mg + \% Fe + \% \text{ Insoluble}$$

LiBr, percent by weight =

100 - Percent (%) Total Impurities

The lithium bromide content shall be as specified in table I.

4.6.2 Physical properties.

4.6.2.1 Typical bulk density. Approximately a 50 g sample of the specimen, weighed to the nearest 0.1 g, shall be transferred to a clean, dry 250 ml glass graduated cylinder. The cylinder shall be inclined and rotated until the entire sample flows freely; care shall be taken to avoid any loss. Thereafter, the surface of the sample shall be leveled by tipping the graduated cylinder slightly forwarded and backward, holding it vertical, without tapping or jarring. The volume occupied by the lithium bromide will be determined in ml. The ratio of the mass (in grams) to the volume (in milliliters) will be the Bulk Density.



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$$\text{Bulk Density} = \frac{\text{Mass of lithium bromide in cylinder (g)}}{\text{Volume occupied in lithium bromide (ml)}}$$

The bulk density value will be in accordance with 3.2.4.1.

4.6.2.2 Optional typical bulk density. Weigh the 100 ml graduated cylinder to the nearest 0.1 g and then add water to the 10 ml mark and reweigh. Add water to the 20 ml mark, reweigh and so forth until the 100 ml mark is reached. The cumulative weight of water added to each 10 ml increment represents the actual volume at that mark, to be used to calibrate the cylinder. For example, if the cumulative weight of water at the 40 ml mark is 39.5 g, the actual volume at 40 ml mark is 39.5 g, and the correction factor (CF) at the 40 ml mark is + 0.5 ml, which must be subtracted from the measured value of 40 ml to give the actual value of 39.5 ml. Repeat the calibration and use the average of the two calibrations.

Weigh 25 g of sample to the nearest 0.1 g and carefully transfer to the dry calibrated 100 ml graduated cylinder attached to the Tap-Pak Volumeter. Allow the sample to slide down the side of the graduated cylinder so it fills the vessel loosely. Read the volume of graduated cylinder (Vl) to the nearest 0.5 ml.

$$\text{Bulk density, (g/ml)} = \frac{\text{Sample weight (g)}}{\text{(Vl - CF)}}$$

Report results to the nearest 0.01 g/ml.

4.7 Inspection test procedures for type II formulation.

4.7.1 Chemical properties.

4.7.1.1 Calcium and magnesium content. The calcium and magnesium content shall be determined by the atomic absorption spectrophotometric method in accordance with ASTM D 511, method B. The calcium and magnesium concentrations shall be as specified in table II.

4.7.1.2 Iron content. The iron content shall be determined by the atomic absorption spectrophotometric method in accordance with ASTM D 1068. The iron concentration shall be as specified in table II.

4.7.1.3 Sulfate content. The sulfate content shall be determined in accordance with ASTM D 516, method A, the gravimetric method. The sulfate content shall be as specified in table II.

4.7.1.4 Instrumental analysis. For Ca, Mg, Fe and $SO_4$ using Inductively Coupled Plasma/Atomic Emission Spectrometry (ICP). The LiBr materials are analyzed by ICP using matrix matched samples and standards containing an internal standard solution. The instrument should be operated at 1200 Watts forward power. The following analytical lines should be used:



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| | |
|---|---|
| Ca - 3933.66 - Analyte | Cd - 2265.10 - Internal Standard |
| Mg - 2790 79 - Analyte | Y - 3710.30 - Internal Standard |
| Fe - 2599.40 - Analyte | |
| S - 1820.40 - Analyte | |

Table V shows analyte and internal standard pair ratios for type II formulation:

Table V. Analyte and internal standard pair ratios (type II formulation).

| | |
|---|---|
| Ca to Y | Fe to Cd |
| Mg to Y | S to Cd |

Samples and standards are adjusted to contain 0.5 g of Li/100 ml and 10 ppm Y and Cd. Samples should be prepared by weighing 11.8 g of LiBr (brine) (52-54 percent) and transferring to a 100 ml volumetric flask containing the Y and Cd internal standard plus 1.0 ml of concentrated HCl. A blank is prepared using the matrix matching material containing 0.5 g of Li, the internal standards and 1 ml of concentrated HCL.

$$\text{Analyte, ppm} = \frac{\text{ICP analyte [mg/l] - ICP blank [mg/l] x 100}}{\text{Sample weight (g)}}$$

4.7.1.5 Lithium bromide content. Determine the bromide ion concentration in the sample by the gravimetric method. The bromide ion content is weighed as silver bromide and then calculated as an equivalent amount of lithium bromide. Pipette 10 ml of the type II formulation sample at 25 +/- 0.1°C (77 + 1°F) into a liter volumetric flask and add distilled water to the 1 liter mark. Shake the solution to mix it thoroughly. Pipette 50 ml of this solution into a 400 ml beaker. Add 200 ml of distilled water to the beaker. While stirring, gradually add a 5 percent solution of silver nitrate, acidified with 1.0 ml of nitric acid, until the precipitate coagulates. Heat the solution to 60°C (140°F) and let the precipitate settle; then add silver nitrate solution one drop at a time until the precipitation is complete. Set the flask aside for at least two hours, or preferably overnight. Pour the solution through a tared Gooch crucible. Wash all the precipitate from the beaker into the crucible with 0.05 g/l silver nitrate solution. Wash the precipitate with the minimum amount of 1:99 nitric acid needed to remove excess silver nitrate, and then wash twice with distilled water to remove the acid. Dry the crucible and its contents at 100°C (212°F) for one hour, then at 140°C (285°F) for another hour. Cool in a desiccator and weigh. Repeat heating and drying until successive weighings agree within 0.5 mg. Calculate the percent by weight of lithium bromide as follows:

$$\text{Lithium Bromide, percent by weight} = \frac{46.25 \times \text{weight of precipitate (g)}}{0.5 \text{ ml} \times \text{Specific gravity of sample at } 25^\circ\text{C } (77^\circ\text{F})}$$



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4.7.1.5.1 Optional lithium bromide content. Take a 300 ml sample and adjust the temperature to either 25°C or 30°C, whichever is the nearest to room temperature. Transfer sample to a hydrometer jar. By means of a precision hydrometer (giving Specific Gravity, 60°F/60°F) obtain the reading. For accurate results the hydrometer must be free grease films. Use cleaning solution if necessary, particularly on the stem. Deduct 0.002 from the hydrometer reading to obtain the brine density in g/ml. Using brine density value obtained above and the adjusted temperature, refer to the plot of density versus weight percent LiBr, as shown in figure 1, to obtain the percent by weight of LiBr in the sample. The lithium bromide content shall be as specified in table II.

4.7.1.6 Alkalinity. The alkalinity of the sample shall not be greater than 0.105 N and not less than 0.09 N. The alkalinity shall be determined by a color-change titration using phenolphthalein as the indicator. Pipette 5 ml of the solution into a white casserole. Add two drops of phenolphthalein indicator and stir the solution. While stirring, slowly titrate 0.02 N hydrochloric acid from a burette until the red color fades completely. Calculate the sample's alkalinity as follows:

$$\text{Alkalinity of sample (N)} = \frac{\text{Milliliters of acid used} \times 0.02\ \text{N}}{5\ \text{ml sample}}$$

The alkalinity shall be as specified in table II.

4.7.1.7 Water insoluble. Vigorously shake the sample to insure that undissolved matter is suspended in solution. Immediately measure out 125 ml of the sample using a 250 ml graduated cylinder. Filter the 125 ml of solution through a 0.45 micron millipore filter, using distilled water to rinse undissolved matter from the cylinder to the filter. Continue washing the suspended matter until a white precipitate appears. Use about 25 ml of distilled water from a wash bottle to transfer solids from the filter to a clean 100 ml beaker, previously dried at 105°C (220°F) for two hours, cool the beaker in a desiccator and weigh. Repeat until successive weighings agree within 0.5 mg. Determine percent by weight of water insoluble as follows:

$$\text{Water insoluble, percent by weight} = \frac{[(\text{Mass of beaker + solids})\ (\text{g}) - (\text{Mass of beaker})\ (\text{g})] \times 100}{125\ \text{ml} \times \text{Specific gravity of sample at } 25°\text{C } (77°\text{F})}$$

The water insoluble (suspended solids) shall be as specified in table II.

4.7.2 Physical properties.

4.7.2.1 Specific gravity. A clean, dry 25 ml pycnometer shall be weighed. Mass shall be recorded to the nearest milligram. The stopper shall be removed and the pycnometer filled with recently boiled distilled water that has been cooled. The stopper shall be inserted, any excess water removed, and the pycnometer allowed to reach 25°C (77°F). The pycnometer shall be weighed, and the mass of



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the water and pycnometer recorded to the nearest milligram. The procedure shall be repeated using type II formulation lithium bromide. The specific gravity of the type II formulation shall be calculated as follows:

$$\text{Specific gravity} = \frac{[(\text{Mass of type II formulation and pycnometer}) (g) - (\text{Mass of pycnometer}) (g)]}{[(\text{Mass of water and pycnometer}) (g) - (\text{Mass of pycnometer}) (g)]}$$

4.8 Inspection of packaging. Sample packs, the inspection of the packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

5.1 Preservation. Not applicable.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Type I formulation. Unit content quantity shall be 100 pounds (45.4 kilograms). The container(s) (unit pack) shall be as specified for the level of packing.

5.2.1.1 Level A.

5.2.1.1.1 Pails. Pails shall conform to PPP-P-704, type II, Class 5 or 11 at the suppliers option. Pails and covers shall be provided with an interior epoxy coating which shall not affect nor be affected by the contents. Pail closure shall be in accordance with the appendix to the pail specification.

5.2.1.2 Level B.

5.2.1.2.1 Pails. Pails shall conform to level A.

5.2.1.2.2 Drums. Fiber drums shall conform to PPP-D-723, type III, grade A, class 1 or 2 at the supplier's option. Drums shall be provided with plastic, heat-sealed, bag-type liners conforming to MIL-L-10547, type IV, grade E or F at the supplier's option. The liners shall not affect or be affected by the contents.

5.2.1.3 Level C.

5.2.1.3.1 Pails. Pails shall conform to level A.

5.2.1.3.2 Drums. Drums shall be as specified for level B except that type II shall apply.

5.2.2 Type II formulation. Unit content quantity shall be 395 pounds (179.6 kilograms).

5.2.2.1 Levels A, B, and C. Containers shall conform to MIL-D-43703, and unless otherwise specified size 3 (30 gallon).



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5.3 Palletization. When specified (see 6.2), filled unit packs shall be palletized in accordance with MIL-STD-147 and the applicable container specification or appendix thereto.

5.4 Marking.

5.4.1 Levels A, B, and C. In addition to any special marking (see 6.2) and herein, unit containers (pails and drums) and palletized unit loads shall be marked in accordance with MIL-STD-129 and the applicable container specification or appendix thereto. Marking information shall include bar coding.

5.4.1.1 Special marking. Container markings shall include the necessary information and warning instructions on product use, handling, transportation, stowage, and disposal including label(s) as required to be furnished on the MSDS (see 3.3.4).

5.5 Material safety data sheet. A copy of the material safety data sheet shall be attached to the shipping document for each destination (see 3.3.4 and 6.5).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Lithium bromide will be used in absorber solutions in lithium bromide air conditioning plants.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, revision and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. First article inspection (see 3.1).
e. Level of packing required (see 5.2).
f. Container "class" if other than supplier's option (see 5.2.1.1.1).
g. Bag-type liner if other than supplier's option (see 5.2.1.2.2).
h. Palletization, when required (see 5.3).
i. Special marking required (see 5.4.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contact. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To insure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirements for a DD Form 1423.



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| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| 4.4 | DI-NDTI-80809 | Test Reports | ---- |

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the items should be first article samples. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Subject term (key word) listing.

Air conditioning
Plants, air conditioning

Preparing activity:
Navy - SH
(Project 6810-N013)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-L-24773(SH) | 24 MARCH 1992 |

**3. DOCUMENT TITLE**
LITHIUM BROMIDE ANHYDROUS AND LITHIUM BROMIDE BRINE

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME TECHNICAL POINT OF CONTACT (TPOC):<br>MR. M. HAMEL (PMS 390)<br>PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS: | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>TPOC: 703-746-3348 | (2) AUTOVON<br>8-332-3488 |
|---|---|---|
| c. ADDRESS (Include Zip Code)<br>COMMANDER, NAVAL SEA SYSTEMS COMMAND<br>SEA 55Z3, DEPARTMENT OF THE NAVY<br>WASHINGTON, DC 20362-5101 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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